K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

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Project Title

Pre-Final Year Project Progress

Submitted by

ROLL NO	NAME
2501060043	Raghav Gupta
2501060061	Prakriti Yadav
2501060066	Dev Sehgal
2501060071	Kanishk Grover
2501060152	Aarohi Kalsi

Faculty Mentor: Mr. Digvijay Singh

Project Overview

In the field of computer applications, data analysis plays an important role in extracting meaningful information from large datasets. Biological health datasets contain structured data related to diseases and population health, that can be effectively analyzed using computational tools.

This project aims to apply basic computing and data analysis techniques to biological health datasets in order to identify disease-related patterns. The focus of the study is on handling, processing, and analyzing health data using programming and statistical methods.

The project handles secondary biological health data and employs data cleaning, organization, visualization, and interpretation through computational approaches.

Specific Objectives

- To apply programming and data analysis concepts to real-world biological datasets
- To collect and manage biological health data using computational tools
- To perform exploratory data analysis using statistical and visualization techniques
- To identify disease-related patterns through data interpretation
- To improve practical understanding of data handling and analysis using programming

Key Features

The project focuses on applying computational techniques to analyze biological health datasets through structured exploratory data analysis. It includes data preprocessing methods such as cleaning, normalization, and handling missing values to ensure reliable analysis.

Statistical summaries and visualizations are generated to reveal disease-related trends and patterns. The system emphasizes data organization, aggregation, and interpretation using programming libraries. The project also promotes reproducibility and documentation of analytical workflows, allowing systematic evaluation of biological health information.

Progress Summary

1. An appropriate Health dataset was chosen from valid government provided datasets from ndap.niti.gov.in.
2. The Health dataset featured various relevant columns – Country, State, Year, Sex, Type Of Disease, Cases and Deaths.
3. The dataset was imported from the website into a Pandas program for cleaning.
4. The dataset was cleaned and saved for further analysis.

Detailed Progress – Development Phase

We decided upon a healthcare related dataset because of its various factors and complex nature.
The dataset was sourced from the National Data and Analytics Platform (NDAP).
The dataset features columns for Country, State, Year, Sex, Type Of Disease, Cases and Deaths.
The dataset was cleaned using a Pandas and numpy program.
- Column names were simplified
- Missing values were filled using mean techniques

Methodology

Dataset was imported to VS-Code.

MINOR PROJECT

- clean.py
- Health Dataset Cleaned.csv
- Health Dataset.csv

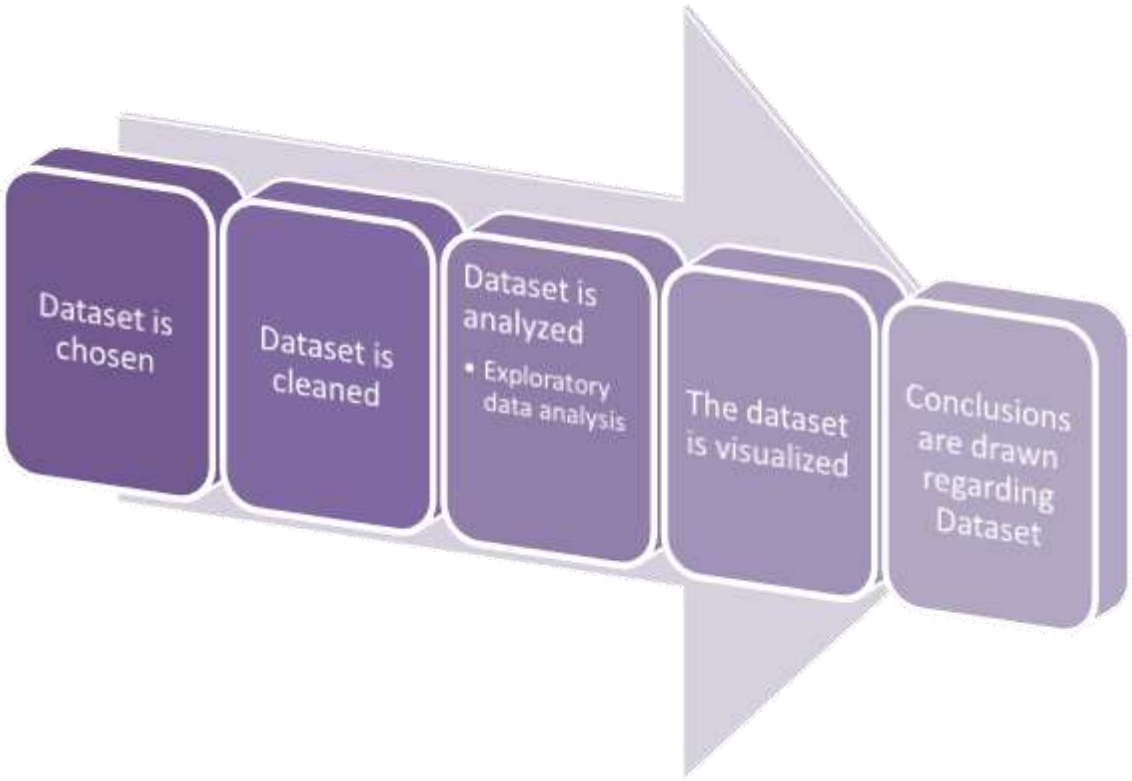
Raw data was displayed.

```python
import pandas as pd
import numpy as np

df = pd.read_csv('Health Dataset.csv')

print("original dataset info:")
print(f"shape: {df.shape}")
print(f"missing values:\n{df.isnull().sum()}\n")
```

Column Names were simplified and missing values were filled in using mean

```python
# simplify column names

newColumnNames = {
    'Country': 'Country',
    'State': 'State',
    'Year': 'Year',
    'Gender': 'Gender',
    'Type Of Disease': 'DiseaseType',
    'Cases (UOM:Number), Scaling Factor:1': 'Cases',
    'Deaths (UOM:Number), Scaling Factor:1': 'Deaths'
}

df.rename(columns=newColumnNames, inplace=True)
print("column names changed:")
print(df.columns.tolist())


# fill missing values in gender

df = df.dropna(subset=['Gender'])
print(f"\nafter removing rows with missing gender: {df.shape}")



# fill missing values in cases

if df['Cases'].isnull().sum() > 0:
    cases_mean = df['Cases'].mean()
    df['Cases'] = df['Cases'].fillna(cases_mean)
    print(f"filled missing values with mean: {cases_mean:.2f}")

# fill missing values in deaths

if df['Deaths'].isnull().sum() > 0:
    deaths_mean = df['Deaths'].mean()
    df['Deaths'] = df['Deaths'].fillna(deaths_mean)
    print(f"filled missing values with mean: {deaths_mean:.2f}")


# verify data

print(f"\ncleaned dataset info:")
print(f"shape: {df.shape}")
print(f"missing values:\n{df.isnull().sum()}\n")
print("first few rows of cleaned data:")
print(df.head(10))

# save the cleaned dataset

output_file = 'Health Dataset Cleaned.csv'
df.to_csv(output_file, index=False)
print(f"\ncleaned dataset saved to: {output_file}")
```

Prototype/Project Flow diagram/Architecture



Project Timeline

- Problem statement was chosen.
- Problem statement was approved.
- Synopsis Presentation was created.
- Synopsis Report was finalized.
- Dataset was chosen.
- Dataset was cleaned.
- Midsem Presentation was created.

Next Steps and Goals

- Dataset will be analyzed.
- Dataset will be visualized using matplotlib and streamlit.
- Conclusions regarding the dataset will be drawn.
- Final report will be made.

Data & Resources

- https://ndap.niti.gov.in
- https://pandas.pydata.org/docs/
- https://numpy.org/doc/
- https://projexa.ai/student/e-resources

Project Usecases & Scope

The scope of this project is limited to exploratory analysis of publicly available biological health datasets using computing tools. It covers data acquisition, preprocessing, statistical exploration, and visualization to identify meaningful disease-related insights.
The project does not attempt predictive modeling or clinical decision-making. Instead, it concentrates on demonstrating how computational methods can be applied to structured health datasets to enhance understanding of biological trends and patterns. The scope ensures practical implementation within academic constraints while maintaining analytical rigor.

Expected Results & Impact

The project is expected to produce meaningful statistical summaries and visual insights that highlight patterns within biological health datasets. These findings will demonstrate how computational techniques can simplify data interpretation and reveal disease-related trends.

The expected outcome is improved understanding of data handling, visualization, and analytical reasoning using programming tools. The results will showcase the practical role of computing in biological data exploration.

THANK YOU